

ARTICLE 1: Harvard and MIT Research Networks Comparison

Rethinking Research Excellence: why interdisciplinary teams outperform traditional approaches at Harvard and MIT

12-word headline

Evidence-based research shows interdisciplinary academic teams within Harvard and MIT dramatically outperform the others in terms of knowledge advancement

One-sentence provocative hook

Are Harvard and MIT fully leveraging cross-disciplinary potential?

150-word pitch summary

The Boston area knowledge economy ecosystem has been at the helm of national and global innovation for a long time. A dense network of universities and research centers coexist in vicinity with a vibrant startup and innovative firm ecosystem, hence benefitting from knowledge flows, fruitful exchange of ideas, and prosperity creation. Chief amongst them, Harvard University and the Massachusetts Institute of Technology are the top contributors to cutting-edge research, whilst often excelling at technology transfer efforts. However, recent research on academic collaboration networks from both universities illustrates that Harvard and MIT can channel and capitalize the latent potential of their researchers by facilitating mission-oriented cross-disciplinary academic research. The combination of time series analysis of publications by field of expertise and collaboration networks clearly reveals that the most dynamic contributors to knowledge advancement tend to conduct research efforts that aspire to bridge the gap between disciplines, operating in the interstitial areas between well-known disciplines, by combining novel hypotheses with rigorous research methods, whereas the more narrowly defined and well-trodden research endeavours tend to suffer from diminishing returns and incremental advances, if not directly stagnating in terms of knowledge advancement.

900-word op-ed version

What makes a successful innovation ecosystem?

What separates innovation ecosystems that consistently push the boundaries of knowledge from those that plateau? In the Boston area—one of the world’s most enduring hubs of discovery—the answer lies not only in the presence of elite institutions, but in how they work together.

A data-driven analysis of research collaboration networks across Harvard University, the Massachusetts Institute of Technology, and their affiliated centers reveals a more precise picture. By mapping co-authorship patterns across schools, departments, and teams, the analysis uncovers where collaboration drives outsized impact—and where it falls short.

The results challenge a common assumption: that concentration of talent alone sustains innovation leadership. Instead, Boston’s advantage is rooted in connectivity—particularly when researchers cross disciplinary boundaries and organize around shared problems. At the same time, the data highlights persistent gaps, including siloed clusters and underutilized connections that limit the system’s full potential.

For leaders and policymakers, the implications are immediate. The same mechanisms that have made Boston a global benchmark can, if left unmanaged, become sources of friction.

Understanding how collaboration actually functionsâ€”not just how it is intendedâ€”offers a practical foundation for building more adaptive, high-performing innovation ecosystems.

A comparative analysis between the Harvard and MIT research ecosystems

To understand how collaboration shapes innovation outcomes, we examine the structure of research interactions at a granular levelâ€”mapping how individual researchers connect across institutional and disciplinary boundaries. Using a network-based approach, each researcher is represented as a node, with size reflecting academic output and color indicating school affiliation. The spatial relationships between nodes capture the frequency and intensity of collaboration: the closer the nodes, the stronger the interaction.

This lens reveals more than just patterns of co-authorship. It exposes the underlying topology of knowledge creationâ€”highlighting which fields are tightly interconnected, which remain isolated, and which act as bridges across domains. Clusters that sit at the center of the network tend to be more deeply integrated with other fields, amplifying their influence and capacity to generate spillover effects.

What emerges is a consistent and consequential pattern. The most dynamic areas of research are not confined within traditional disciplinary boundaries but instead thrive at their intersections. These interstitial spacesâ€”where distinct domains overlapâ€”prove disproportionately productive, as they combine diverse perspectives with methodological rigor. Researchers operating in these zones are better positioned to challenge assumptions, recombine ideas, and unlock new avenues of inquiry.

By contrast, more insular and narrowly defined research trajectories often exhibit diminishing returns over time. While they may produce steady output, their contributions tend to be incremental, reflecting the limits of working within established paradigms.

The goal of this analysis, therefore, is not simply to describe collaboration patterns, but to identify the conditions under which they translate into meaningful advances. In doing so, it offers a clearer framework for fostering research environments that balance depth with cross-disciplinary reachâ€”an essential ingredient for sustained innovation and impact for the society.

Harvard

- **2,812 core researchers**
- **7,684 extended researchers (core + collabs)**
- **11 schools, 53 departments**
- **Publications: see visual**
- **Emphasis: medical research, public health, engineering, computer science, business, law, government, urbantech**

MIT

- **1,224 core researchers**
- **4,729 extended researchers (core + collabs)**
- **7 schools, 40 departments**
- **Publications: see visual**
- **Emphasis: computer science, engineering, robotics, basic sciences, business, and life sciences through affiliated labs**

Strengths and Weaknesses: A Tale of Two Institutions

A comparative analysis of research networks at Harvard University and the Massachusetts Institute of Technology reveals two distinct—yet highly complementary—models of innovation. Both institutions anchor the Boston ecosystem, but they do so with different structures, scales, and areas of emphasis.

Harvard operates at a larger institutional scale, with more than 2,800 core researchers and an extended collaboration network approaching 7,700 scholars. Spanning 11 schools and over 50 departments, its research footprint is both broad and deeply embedded across disciplines. Its strengths are particularly evident in fields such as medical research, public health, law, government, and business, alongside growing capabilities in engineering, computer science, and urban technology. This breadth enables Harvard to act as a wide-reaching knowledge hub, with strong links between academia, policy, and applied societal challenges.

MIT, by contrast, is more compact but highly concentrated. With roughly 1,200 core researchers and an extended network of more than 4,700 collaborators, it operates across 7 schools and 40 departments. Its comparative advantage lies in technical depth—particularly in computer science, engineering, robotics, and the basic sciences—augmented by strong integration with business and life sciences through affiliated laboratories. The result is a tightly interconnected research environment optimized for rapid iteration and technological advancement.

These structural differences translate into distinct collaboration patterns. Harvard's scale and disciplinary diversity support a wide range of connections, but can also introduce fragmentation across fields. MIT's smaller, more integrated network fosters density and cohesion, though potentially at the cost of narrower disciplinary reach.

Taken together, the two institutions illustrate a broader principle: there is no single model for innovation leadership. Scale enables reach, while concentration enables intensity. The challenge—and opportunity—lies in how these attributes evolve over time. Time-series analysis of research output and collaboration patterns suggests that both institutions are increasingly converging around high-growth, interdisciplinary domains, particularly those at the intersection of computational, life, and social sciences.

The next section examines these emerging areas more closely, identifying where the most dynamic and promising fields of knowledge are taking shape—and what they signal for the future of innovation.

Research Strategy Recommendations: a call to action

The evaluation of the collaboration networks of research and innovation at Harvard and MIT reveal that the most dynamic contributors to knowledge advancement tend to conduct research efforts that aspire to bridge the gap between disciplines, operating in the interstitial areas between well-known disciplines, by combining novel hypotheses with rigorous research methods, whereas the more narrowly defined and well-trodden research endeavours tend to suffer from diminishing returns and incremental advances, if not directly stagnating in terms of knowledge advancement

We observe notable differences between both institutions: MIT presents a very strong core built around the Sciences and Engineering. However, when considered the constellation of satellite collaboration labs, the life sciences gain a strong presence and output within the MIT community. Considering that sector is slowing down in investment and yields (scientific and financial), potential opportunity to allocate efforts in other sectors: advanced manufacturing, robotics, AI, energy, product design, urbantech.

Complementarily, at Harvard the Medical/Genetics/Pharma life sciences-related areas are tightly knit and present fertile soil for tech transfer efforts. At Harvard, emerging fields in engineering, computer science, and tech-related sectors present the largest growth potential, however they require physical spaces (Allston) to enable interdisciplinary collaboration where invention, creation, and MVP consolidation can occur, hence allowing to host a stronger startup concentration base, as well as a stable liaison with major corporations and leading innovative firms. The Faculty of Arts and Sciences (FAS) concentrates a good deal of core researchers, but the impact is limited as of today - opportunity to bridge the gap between the best research and practice. Finally, the business, law, and government knowledge areas present a more isolated, less connected, and sophisticated ecosystem generally speaking.

Moving forward, a key driver to strengthen knowledge diffusion and the success of the innovation ecosystem can be the constitution of a series of mission-oriented and social challenge-driven innovation centers, thoughtfully conceived to fill in the missing gaps of the innovation pipeline bridge. In order for those centers to fulfill their goal, they should be strategically located, grouped by knowledge domain, and deliberately target the establishment of stable, carefully crafted industry partnerships, thus permitting a solid liaison with leading tech firms operating in the area. Complementarily, a series of support services such as the ones provided by MIT VMS (Venture Mentoring Services) can help strengthen startup consolidation by providing the necessary guidance to successfully cross the bridge of innovation and produce products and services at scale, hence allowing to create prosperity and quality employment opportunities, benefitting the broader society.

ADD

Aggregate Core network of research collaborations at MIT between 2000 and 2026

Current Core network of research collaborations at MIT between 2024 and 2026

Core network of research collaborations at Harvard between 2024 and 2026

Extended network of research collaborations at Harvard between 2024 and 2026

Potential synergies between Harvard, MIT, Mass General and the Longwood Medical Area

Harvard Allston Cambridge Campuses by Schools - Departments no labels.png

Harvard campus spaces by domain of knowledge

MIT Campus by Schools - Departments no labels.png

MIT campus spaces by domain of knowledge

Harvard Longwood Campus by Schools - Departments no labels.png

Longwood campus spaces by domain of knowledge